

SILVIA ARELLANO GARCÍA

Deep Learning · Generative AI · Animation · 3D Computer Vision

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SUMMARY

- First year **PhD Candidate in AI for 3D Gesture Motion** with a focus on deep generative models for human-centric animation and interactive digital characters.
- **Industry Research Background** at **Electronic Arts** and **Dolby Laboratories**, and **academic research experience** at Universitat Politècnica de Catalunya (UPC) and KTH Royal Institute of technology.
- **Experience in Generative AI** including Diffusion Models, Flow Matching, and Autoregressive Transformers applied to 3D terrains, motion in-betweening, and audiovisual signals.

EDUCATION

PhD in Artificial Intelligence (3D Gesture Motion)

KTH Royal Institute of Technology (KTH)

Feb. 2026 – Present

Stockholm, Sweden

- Expected Focus: Deep Learning for Computer Animation and Human Motion Synthesis.
- Supervised by Prof. Gustav Eje Henter, Prof. Jonas Beskow and Prof. Hedvig Kjellström.

MSc in Machine Learning

Kungliga Tekniska Högskolan (KTH)

Aug. 2023 – June 2025

Stockholm, Sweden

- GPA: 4.83/5.0
- Relevant subjects: Advanced Machine Learning, Advanced Deep Learning, AI for Multiagent Systems, Visualization, Computer Vision.
- Thesis: **Multi-View 3D Asset Generation: A Latent Space Manipulation Approach**

Bachelor Thesis

Aalto University (ERASMUS+ Program)

Jan. 2023 – June 2023

Espoo, Finland

- Grade: 10.0/10.0
- Conducted at the Aalto Mobile Robotics Lab under the supervision of Tomasz Kuckner and Beatriz Otero Calviño.
- Thesis: **A 3D Terrain Generator: Enhancing robotics simulations with GANs.**

BSc in Electrical Engineering (8.24/10.0)

Universitat Politècnica de Catalunya (UPC)

Sept. 2019 – June 2023

Barcelona, Spain

- GPA: 8.24/10.0. Ranked 7th out of 122 students.
- Specialization in Audiovisual Signal Processing.

WORK AND RESEARCH EXPERIENCE

EA Sports

Machine Learning Engineer

Sept. 2025 – Jan. 2026

Stockholm, Sweden

- Fine-tuned pose estimation models for multiple sports applications.
- Designed and executed data collection pipelines and performed data preprocessing.

SEED - Electronic Arts

MSc Thesis Intern

Jan. 2025 – June 2025

Stockholm, Sweden

- 3D asset generation from multi-view images using flow matching models. Built on **TRELLIS** to develop new latent space conditioning strategies, improving geometry and texture consistency. Supervised by Héctor Anadón León and Dr. Judith Bütepage.

Dolby Laboratories

Research intern in AI and acoustics

June 2024 – Aug. 2024

Barcelona, Spain

- Generation of Room Impulse Responses using deep neural networks conditioned on acoustic parameters, developing models like autoregressive transformers, flow matching, and MaskGIT to achieve state-of-the-art performance. Supervised by Dr. Daniel Arteaga.

KTH Royal Institute of Technology

Teaching Assistant

Spring and Autumn semester 2024

Stockholm, Sweden

- Teaching assistant for the course DD2257 Visualization (Period 1) and DD1420 Foundations of Machine Learning (Period 3), responsible for grading assignments and conducting help sessions.

Computer Architecture Department - Universitat Politècnica de Catalunya

Undergraduate Research Fellow

March 2022 – June 2023

Barcelona, Spain

- Exploration of the use of Diffusion Models and Generative Adversarial Networks (GANs) in the process of creation of digital graphics. Supervised by Dr. Beatriz Otero.

- Research about technologies and components used for implementing building blocks in a data space.

PUBLICATIONS

- **Room Impulse Response Generation Conditioned on Acoustic Parameters**
Silvia Arellano, Chunghsin Yeh, Gautam Bhattacharya, Daniel Arteaga. *WASPAA 2025*.
- **A 3D Terrain Generator: Enhancing Robotics Simulations with GANs**
Silvia Arellano, Beatriz Otero, Tomasz Piotr Kucner, and Ramon Canal. *LOD 2023*.
- **Exploring Image Transformations with Diffusion Models: A Survey of Applications and Implementation Code**
Silvia Arellano, Beatriz Otero, Ruben Tous. *LOD 2023*.

PROJECTS

- Motion In-betweening with Diffusion Models** Aug. 2024 – Dec. 2024
Advanced Individual Course in CS Stockholm, Sweden
- Researched and implemented diffusion-based architectures to address high-fidelity motion synthesis and in-betweening challenges.
- Challenge Based Innovation Project** Sept. 2022 – Dec. 2022
CERN - EU-ATTRACT Academy Barcelona - Geneva
- Developed a hardware-software prototype using C# to track and create digital 3D representations of tumor locations.
 - Collaborated within an international consortium including CERN and ESADE Business School.

AWARDS AND SCHOLARSHIPS

- MOBINT Fellowship** 2023
Government of Catalonia Barcelona, Spain
- Awarded to top-tier students from Catalan universities to support international academic stays.
- Research Collaboration Grant** Oct. 2022
Spanish Ministry of Education Barcelona, Spain
- Competitive grant awarded for conducting specialized research at the UPC Department of Computer Architecture.
- Recognition to the Best Academic Records** Sept. 2021
ETSETB TelecomBCN Barcelona, Spain
- Ranked 6th out of 224 students in the initial engineering module.
- High School Regional Merit Award** Jan. 2018
Government of Aragón Aragón, Spain
- Ranked in the top 15 out of 10,873 students in the region.

LEADERSHIP AND VOLUNTEERING

- She's In The Game - Mentor** 2025
- Mentor for teenage girls in the process of creating their first videogame.
- HackUPC - Marketing Director** 2020 – 2023
- Directed a 5-person team managing communications and branding for Europe's largest student hackathon.
 - Organized large-scale technological activities and workshops for students.
- BitsxLaMarató - Director and Organizer** 2020 – 2022
- Led the organization of a charity hackathon contributing to La Marató de TV3 for biomedical research.
- Saló de l'Ensenyament - Advisor** 2022
- Selected by UPC to promote the BSc of Electrical Engineering amongst high school students.
- Delegació d'Alumnes de Telecomunicacions** Sept. 2019 – Jan. 2023
- Acted as a bridge between the student body, professors, and administration as part of the Student Council.

SKILLS

- **Programming:** Python, C, C++, C#, Matlab, Java, HTML.
- **Frameworks & Libraries:** PyTorch, Scikit-learn, OpenCV, NumPy, TensorFlow.
- **Tools & Graphics:** Git, Unity3D, LaTeX, Blender.
- **Languages:** Spanish and Catalan (Native), English (Advanced, C1), French (Intermediate, B1), Swedish (Beginner, A2).
- **Hobbies:** Canoe slalom, board games, swimming, origami, crochet, Spanish guitar.